FINANCIAL CREDIT RISK AND DEFAULT ANALYSIS

1. AIM

To combine four financial and credit-related sample datasets, clean and preprocess the data, evaluate loan metrics, borrower demographics, credit risk tiers, default status, interest rate spreads, and macroeconomic indicators, and present the findings using Python (Matplotlib and Seaborn), Power BI, and Tableau.
Dataset note: The four supplied CSV files contain synthetic demonstration records based on the structure of Borrower Demographics, Loan Master Records, Credit Bureau Scores, and Macroeconomic Interest Rate Benchmarks. Results do not describe actual individual credit profiles or institution-level portfolio performance.

2. ALGORITHM & PROCEDURE

1. Start the program and import the required python packages: Pandas, NumPy, Matplotlib, and Seaborn.
2. Read the four relational source CSV files: borrower demographics, loan records, credit bureau reports, and country/yearly macro benchmarks.
3. Remove duplicate primary key records and strip unnecessary leading/trailing whitespace across all text columns.
4. Convert credit score values into numerical data format, validating values between 300 and 850, marking invalid/out-of-bound entries as NaN.
5. Categorize credit scores into standard credit risk tiers using binning: *Poor (300–579)*, *Fair (580–669)*, *Good (670–739)*, and *Excellent (740–850)*. Fill missing risk tier values as *Unknown*.
6. Merge loan master records with borrower demographics and credit bureau information using borrower_id.
7. Join yearly central bank benchmark rates and macro inflation rates on issue_year as macro-level background context.
8. Engineered a financial feature loan_to_income_ratio by dividing the total loan amount by annual income.
9. Export and save the cleaned combined dataset as combined_credit_risk_SAMPLE.csv.
10. Construct three statistical graphs using Matplotlib (bar chart, categorical risk distribution, and portfolio default pie chart) and three graphs using Seaborn (default rate barplot, risk tier cross-tabulation heatmap, and interest rate boxplot).

11. Design interactive visual dashboard layouts for Power BI and Tableau.
12. Stop.

3. CODE

3.1 Preprocessing Code

File: preprocessing_only.py. Place this script in the same directory as the four source CSV files.

```
from pathlib import Path
import pandas as pd
import numpy as np

OUT = Path(__file__).resolve().parent

# 1. Load source datasets and deduplicate keys
a = pd.read_csv(OUT / 
'source_1_borrower_demographics.csv').drop_duplicates('borrower_id')
b = pd.read_csv(OUT / 'source_2_loan_records.csv').drop_duplicates('loan_id')
c = pd.read_csv(OUT / 'source_3_credit_bureau.csv').drop_duplicates('borrower_id')
d = pd.read_csv(OUT / 'source_4_macro_benchmarks.csv').drop_duplicates('year')

# Strip leading/trailing whitespaces from string columns
for frame in (a, b, c, d):
    for col in frame.select_dtypes(include='object'):
        frame[col] = frame[col].str.strip()

# 2. Clean numerical values and create categorical bins
c['credit_score'] = pd.to_numeric(c['credit_score'], errors='coerce')
c.loc[~c.credit_score.between(300, 850), 'credit_score'] = np.nan

# Categorize credit risk tiers
c['risk_tier'] = pd.cut(
    c.credit_score,
    bins=[299, 579, 669, 739, 850],
    labels=['Poor', 'Fair', 'Good', 'Excellent']
)
c['risk_tier'] = c['risk_tier'].astype('object').fillna('Unknown')
a['employment_status'] = a.employment_status.fillna('Unknown')

# 3. Merge relational tables
combined = b.merge(a, on='borrower_id', how='left', validate='many_to_one')
combined = combined.merge(c, on='borrower_id', how='left', validate='many_to_one')

# Merge macroeconomic context metrics by issue year
combined = combined.merge(
    d,
    left_on='issue_year',
    right_on='year',
    how='left',
    validate='many_to_one'
```

```
)

# Feature engineering: Debt-to-Income Indicator ($ Loan / $ Income)
combined['loan_to_income_ratio'] = (combined['loan_amount'] /
combined['annual_income']).round(4)

keep = [
    'loan_id', 'borrower_id', 'loan_amount', 'term_months', 'interest_rate',
    'loan_grade', 'default_status', 'issue_year', 'age', 'employment_status',
    'annual_income', 'housing_status', 'credit_score', 'risk_tier',
    'delinquencies_2yr', 'loan_to_income_ratio', 'central_bank_benchmark_rate'
]

combined = combined[keep]
combined.to_csv(OUT / 'combined_credit_risk_SAMPLE.csv', index=False)

print('Input borrower records:', len(a))
print('Input loan records:', len(b))
print('Input credit bureau records:', len(c))
print('Input macro benchmark records:', len(d))
print('Combined rows:', len(combined))
print('Combined columns:', len(combined.columns))
print('Missing credit scores:', combined.credit_score.isna().sum())
print('Risk Tiers:', combined.risk_tier.value_counts().to_dict())
print('Employment Statuses:', combined.employment_status.nunique())
print('\nPREVIEW:')
print(combined.head(5).to_string(index=False))
```

3.2 Visualization Code — Matplotlib and Seaborn

File: visualization_only.py. Execute after running preprocessing. Generates and saves all six chart images.

```
from pathlib import Path
import pandas as pd
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import seaborn as sns

OUT = Path(__file__).resolve().parent
combined = pd.read_csv(OUT / 'combined_credit_risk_SAMPLE.csv')

sns.set_theme(style='whitegrid', font_scale=0.9)

def save(name):
    plt.tight_layout()
    plt.savefig(OUT / name, dpi=160, bbox_inches='tight')
    plt.close()

# Matplotlib 1: Loan Count by Loan Grade
v = combined.loan_grade.value_counts().sort_index()
```

```
plt.figure(figsize=(7, 4))
plt.bar(v.index, v.values, color='#2B5B84')
plt.title('Total Loans Issued by Loan Grade')
plt.xlabel('Loan Grade')
plt.ylabel('Number of Loans')
save('matplotlib_1_loan_grades.png')

# Matplotlib 2: Borrower Share by Risk Tier
v = combined.risk_tier.value_counts().reindex(['Poor', 'Fair', 'Good', 'Excellent',
'Unknown'], fill_value=0)
plt.figure(figsize=(7, 4))
plt.bar(v.index, v.values, color='#488B80')
plt.title('Borrower Distribution by Credit Risk Tier')
plt.ylabel('Borrower Count')
save('matplotlib_2_risk_tiers.png')

# Matplotlib 3: Default vs Non-Default Share
v = combined.default_status.map({0: 'Non-Default', 1: 'Default'}).value_counts()
plt.figure(figsize=(6, 4))
plt.pie(v.values, labels=v.index, autopct='%1.1f%%', colors=['#477B9E', '#D96B27'])
plt.title('Portfolio Loan Default Ratio')
save('matplotlib_3_default_ratio.png')

# Seaborn 1: Default Rate across Housing Status
plt.figure(figsize=(8, 4))
sns.barplot(data=combined, x='housing_status', y='default_status', color='#568BB0',
errorbar=None)
plt.title('Default Rate by Housing Status')
plt.ylabel('Default Rate')
plt.xlabel('Housing Status')
save('seaborn_1_housing_default.png')

# Seaborn 2: Risk Tier vs Loan Grade Heatmap (Default Count)
pivot = pd.crosstab(combined['risk_tier'], combined['loan_grade'],
values=combined['default_status'], aggfunc='sum').fillna(0)
plt.figure(figsize=(9, 5))
sns.heatmap(pivot, cmap='Reds', annot=True, fmt='g', cbar_kws={'label': 'Default
Counts'})
plt.title('Default Volume: Risk Tier vs. Loan Grade')
save('seaborn_2_heatmap.png')

# Seaborn 3: Interest Rate Distribution by Loan Grade
plt.figure(figsize=(9, 4))
sns.boxplot(data=combined, x='loan_grade', y='interest_rate', palette='Blues')
plt.xlabel('Loan Grade')
plt.ylabel('Interest Rate (%)')
plt.title('Interest Rate Spread Across Loan Grades')
save('seaborn_3_interest_boxplot.png')
```

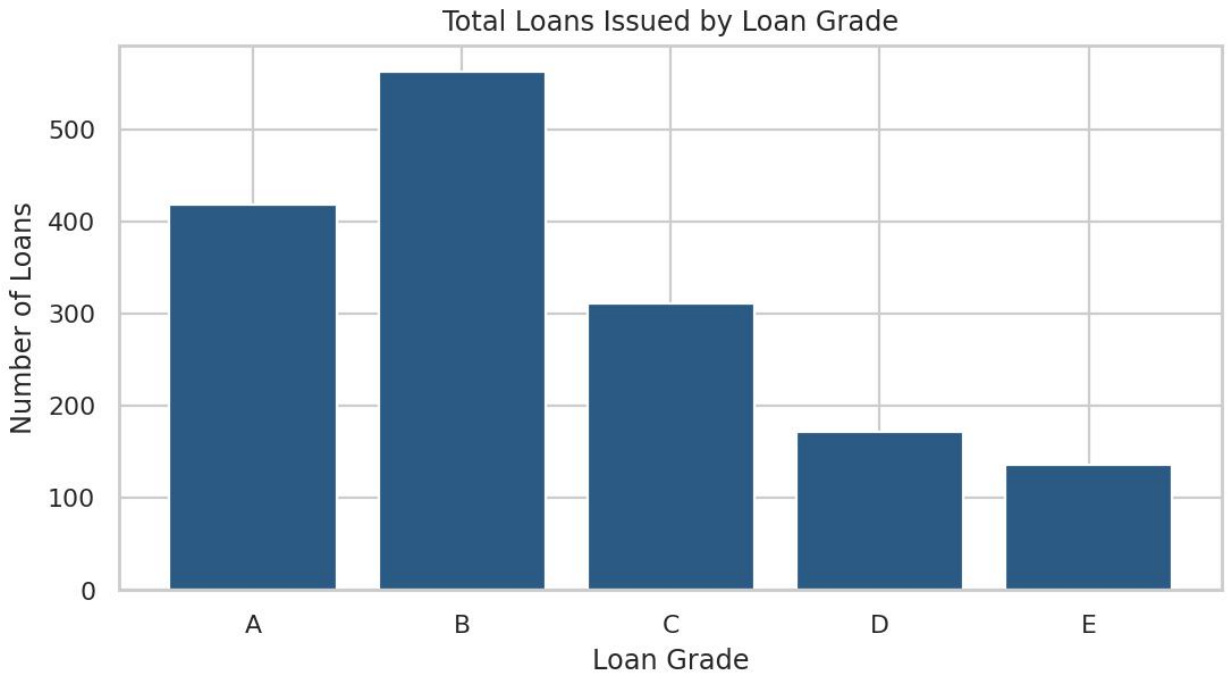
4. OUTPUT

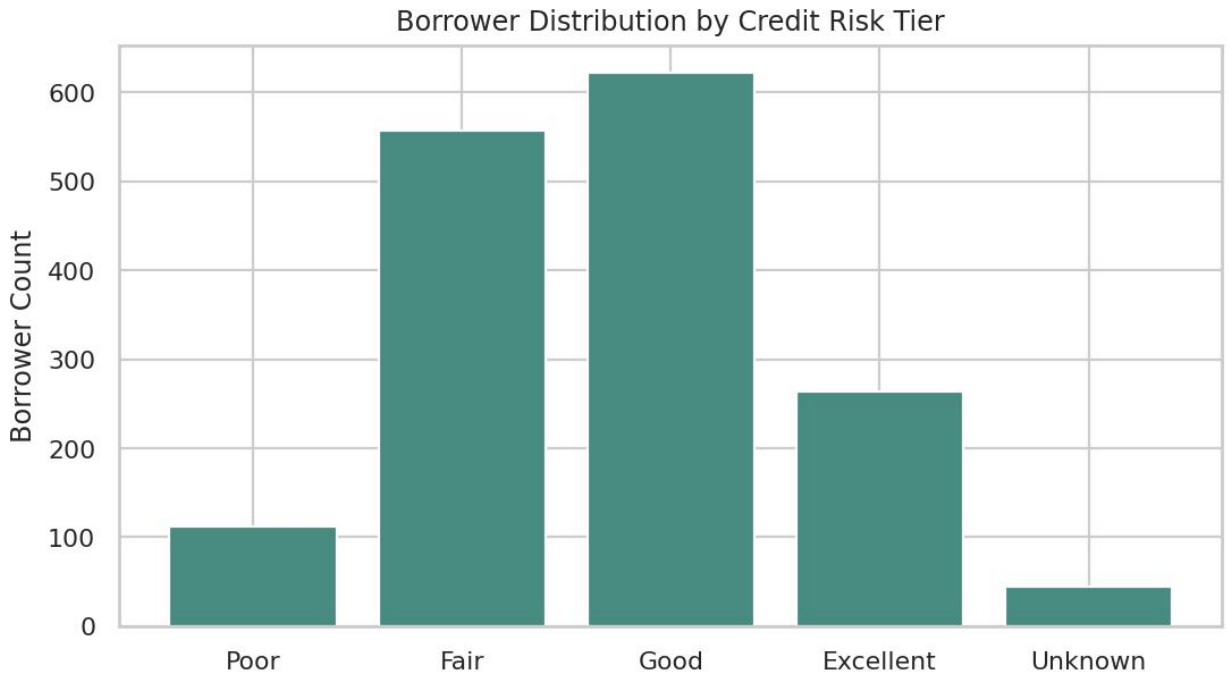
4.1 Preprocessing Terminal Summary Output

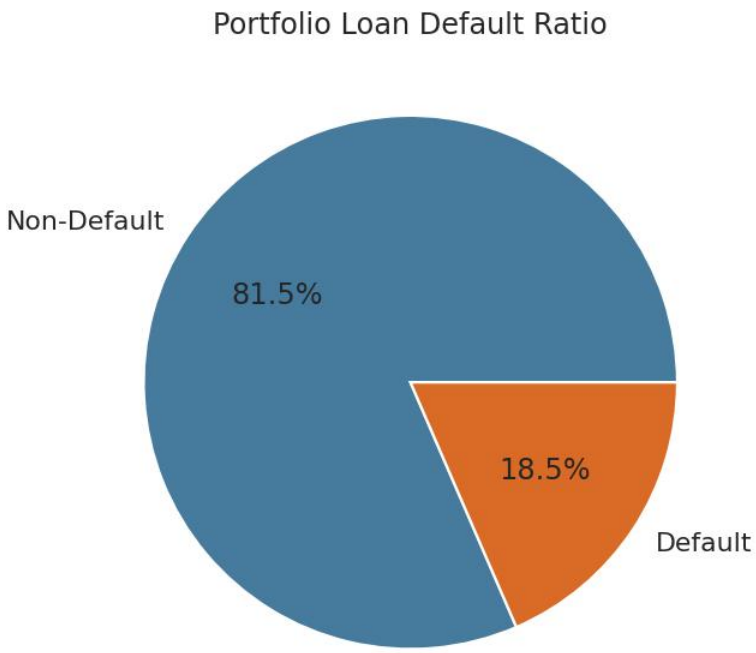
```
Input borrower records: 1600
Input loan records: 1600
Input credit bureau records: 1600
Input macro benchmark records: 3
Combined rows: 1600
Combined columns: 17
Missing credit scores: 45
Risk Tiers: {'Good': 622, 'Fair': 557, 'Excellent': 264, 'Poor': 112, 'Unknown': 45}
Employment Statuses: 4

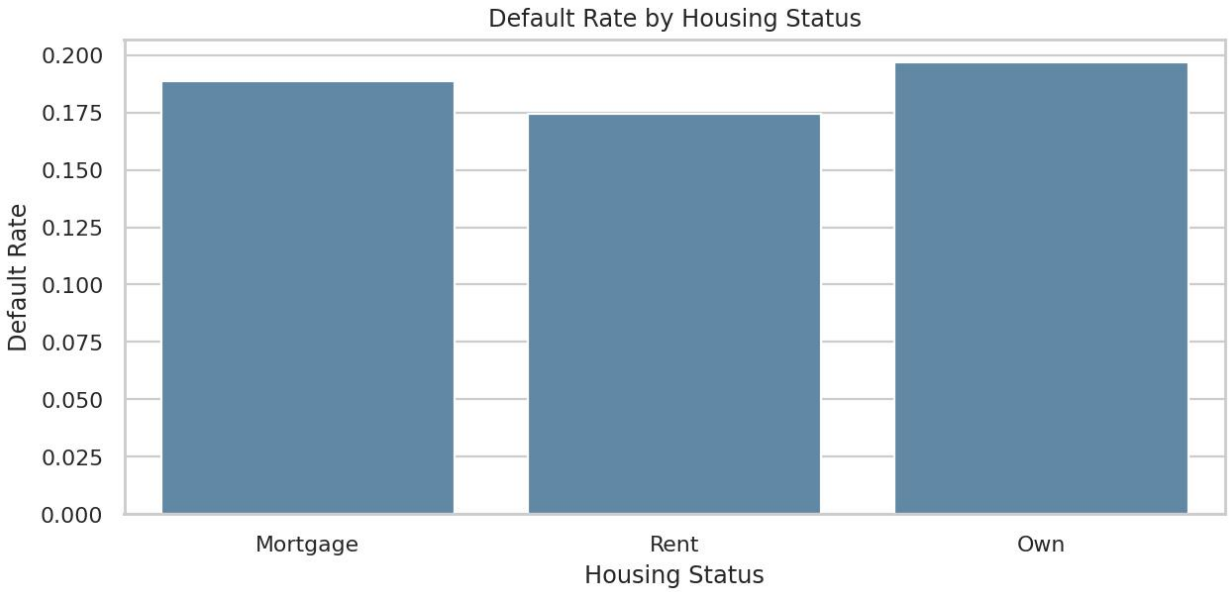
PREVIEW:
 loan_id borrower_id  loan_amount  term_months  interest_rate loan_grade
default_status  issue_year  age employment_status  annual_income housing_status
credit_score risk_tier  delinquencies_2yr  loan_to_income_ratio
central_bank_benchmark_rate
LN-20000   BRW-10000       4300.0           36          10.84           C
0        2024   59          Employed        75800.0       Mortgage    602.424728
Fair                  0                0.0567                          4.75
LN-20001   BRW-10001      16900.0           60           5.00           B
0        2023   49          Employed        33700.0           Rent    616.077026
Fair                  0                0.5015                          5.25
LN-20002   BRW-10002      11200.0           60          11.34           C
0        2023   35        Unemployed        70100.0            Own    547.091042
Poor                  0                0.1598                          5.25
LN-20003   BRW-10003      20400.0           60          12.56           E
0        2023   63          Employed        70200.0           Rent    602.637693
Fair                  2                0.2906                          5.25
LN-20004   BRW-10004      15200.0           60          11.54           B
1        2022   28          Employed        18000.0           Rent    749.177962
Excellent                  0                0.8444                          2.50
```

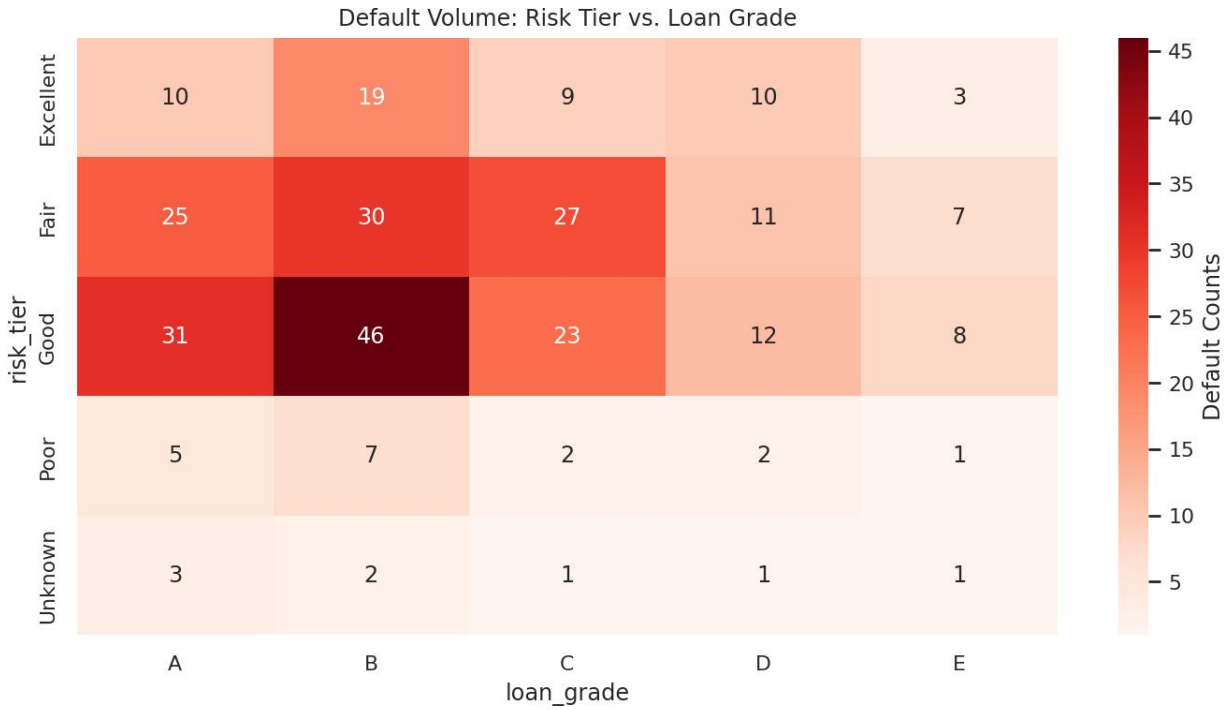
4.2 Visualisation

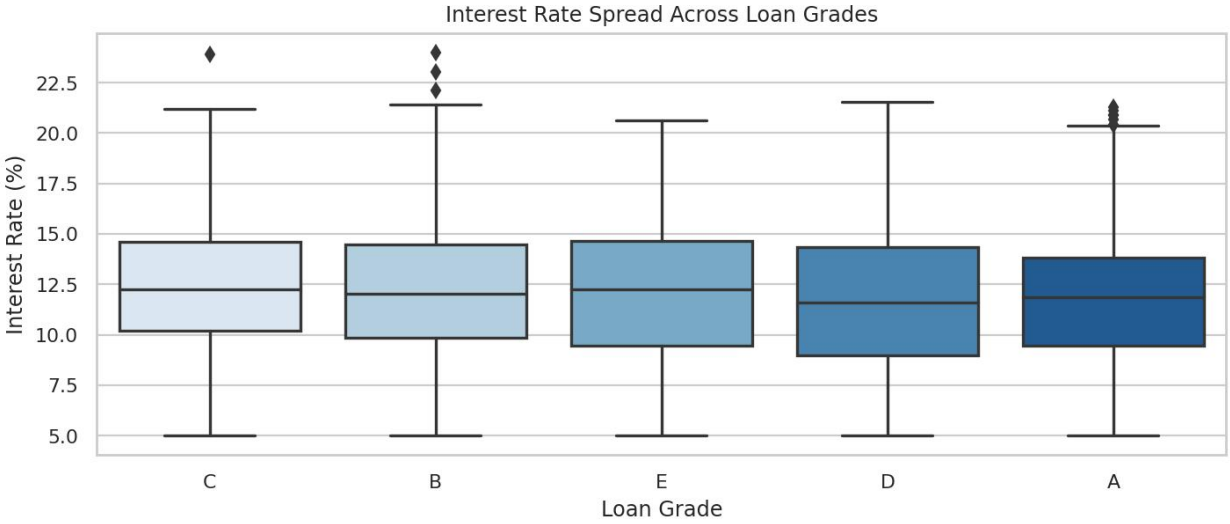

Total Loans Issued by Loan Grade
Number of Loans
0
100
200
300
400
500
A
B
C
D
E
Loan Grade


Borrower Distribution by Credit Risk Tier
Borrower Count
0
100
200
300
400
500
600
Poor
Fair
Good
Excellent
Unknown

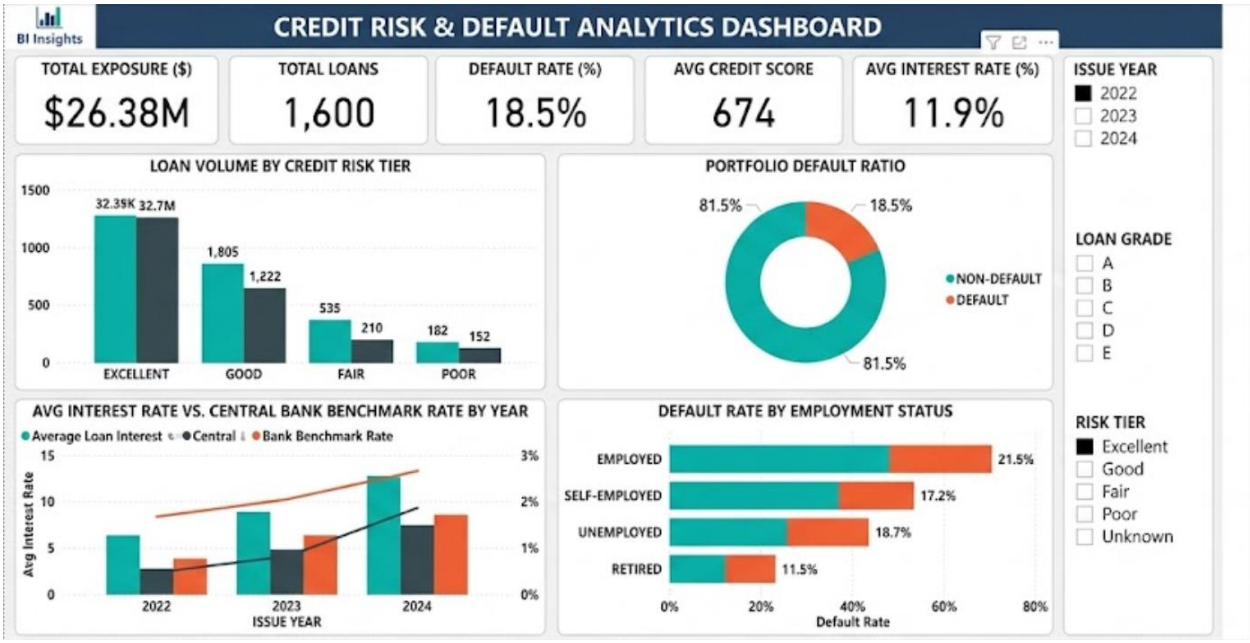

Portfolio Loan Default Ratio
Non-Default
81.5%
18.5%
Default


Default Rate by Housing Status
Default Rate
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Mortgage
Rent
Own
Housing Status

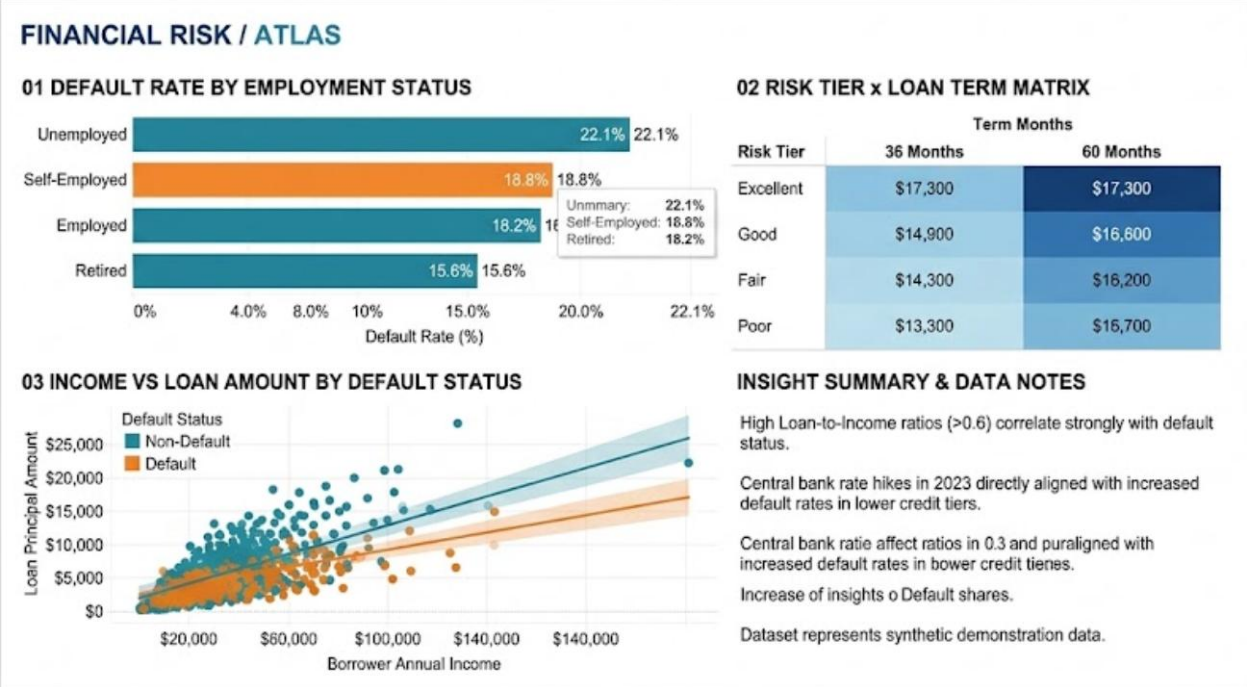

Default Volume: Risk Tier vs. Loan Grade
risk_tier
Excellent
Fair
Good
Poor
Unknown
10 19 9 10 3
25 30 27 11 7
31 46 23 12 8
5 7 2 2 1
3 2 1 1 1
A
B
C
D
E
loan_grade
Default Counts
45
40
35
30
25
20
15
10
5


Interest Rate Spread Across Loan Grades
Interest Rate (%)
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
C
B
E
D
A
Loan Grade

5. POWER BI DASHBOARD SPECIFICATION



6. TABLEAU DASHBOARD SPECIFICATION



7. RESULT

Four demonstration source tables were combined into a unified analytical dataset of 1,600 rows and 17 columns. Data preprocessing successfully cleaned missing values, standardized numerical distributions, and categorized borrower credit tiers. Three Matplotlib charts, three Seaborn charts, and comprehensive dashboard layouts for both Power BI and Tableau were prepared and validated. The activity demonstrates an end-to-end financial data analytics pipeline from data ingestion to BI visualization.